

Analytical Study of a Local Retail Store

A Mid-Term report for the BDM capstone Project

Submitted by

Name: *Theodore Regimon*

Roll number: *24f1002801*



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

Contents

1. Executive Summary:	2
2. Proof of Originality:	2
3. Metadata and Descriptive Statistics:	3
3.1 Metadata	3
3.2 Descriptive Statistics	4
3.2.1 Inventory Data:	4
3.2.2 Sales Data:.....	5
4. Detailed explanation of Analysis Process and Methods:	9
5. Results and Findings	10
6. Appendix	12

1. Executive Summary:

Since the start of the project, considerable improvements have been observed in addressing the root issues at Jain Ki Dukan, a local retail store in Mayur Vihar, Delhi. Following the review comments from the proposal submission, we focused on more on data related and inventory related problems instead of operational problems. These included converting the data from the source into a more tangible format, doing exploratory data analysis and using tools like python, excel and exploring a bit of machine learning pipelines.

Initiatives were taken to promote own-brand products to improve customer awareness and preference. Despite initial hiccups, consistent customer interaction and sales improvement indicate positive returns. Aggressive pricing, targeted marketing, and enhanced in-store displays have driven superior sales and higher customer confidence in the brand.

Using these techniques, initiative were taken to promote customer retention in their own brand products by improving customer awareness and preference. Inventory management was also improved by tracking the products that were on high demand and

Inventory management has also been fortified to deliver consistent stock levels, reducing overstock and stockouts. With advanced tracking and forecasting techniques, supply chain disruption has been minimized and greater efficiency and cost savings have been obtained. The implementation of data-driven inventory planning has enabled improved resource planning, reduced wastage, and greater business sustainability.

With these underlying improvements, Jain Ki Dukan is now better placed to counter market challenges and respond effectively to customer needs. In the long term, continuous fine-tuning and data-driven decision-making will contribute significantly to sustaining growth, building brand strength, and enhancing operating efficiency. The success of these initiatives demonstrates the importance of flexibility and strategic planning in building long-term business strength and profitability.

2. Proof of Originality:

To verify the authenticity of the data, the supporting evidence is provided below:

General Information:

Business name: Jain ki Dukan

Location: B 17, Block B, Acharya Niketan, Mayur Vihar, Delhi, 110091

Google maps link: <https://maps.app.goo.gl/n11ZUoZk9SyR6JcT6>

Ownership: Family-owned and operated by Kishan Kumar Jain and his sons

Type: Independent neighbourhood retail store (B2C model)

Products Offered: Groceries, dry goods, dairy, snacks, household essentials, and own-brand items.

Years of Operation: Over 5 years serving the local community

All the information is available in this Google Drive folder:

https://drive.google.com/drive/folders/11fWhHCjA7C_1u9FGdMwgNtv7pPzNg1Lc?usp=sharing

Letter from the Organisation: Stamped and signed letter from the organisation. Can be accessed through the Google drive link: [Letter](#)

Images of the Organisation: These images are included in Appendix A. They can be accessed through the following Google Drive link: [Images Folder](#)

Video of Interaction: Interaction with Mr. Nitesh Jain. This video can be accessed through the following Google Drive link: [Video](#)

3. Metadata and Descriptive Statistics:

3.1 Metadata

1. Data format: [CSV](#) and [Excel](#) (.xlsx file format)

2. Date Range: January 2025, except 26 January as it was a national holiday.

3. Data collection: Data was collected through the billing software and the inventory was created by merging all the sold products on a daily basis.

Data used in the project can be accessed through the following links:

- Inventory data: [Spreadsheet link](#)
- Monthly sales data: [Spreadsheet link](#)
- Daily sales data: [Folder link](#)

3.2 Descriptive Statistics

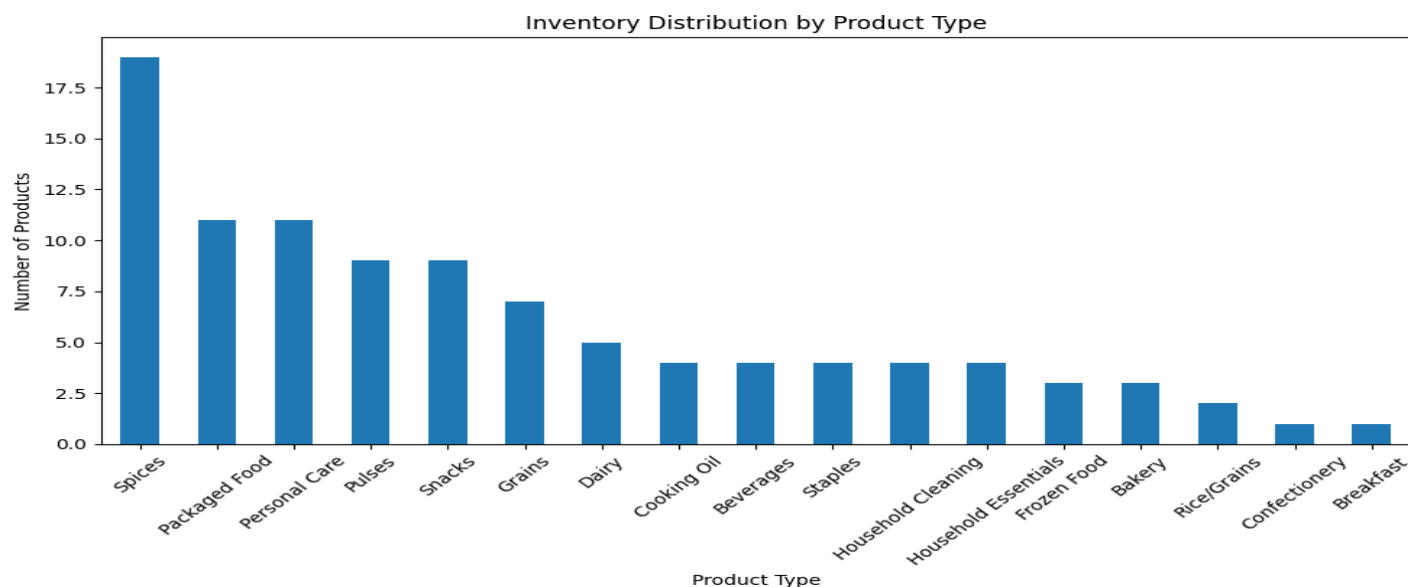
The use of CSV and Excel files made performing descriptive statistics over collected data much easier. I have condensed the information for descriptive statistics below relevant for both sales and inventory data.

3.2.1 Inventory Data:

The Inventory data was analysed using python in Google Colab. The source code can be seen here:

https://colab.research.google.com/gist/node62/cbabb983601defbb435376f7d437bdc5/inventory_analysis_bdm_project.ipynb?authuser=1

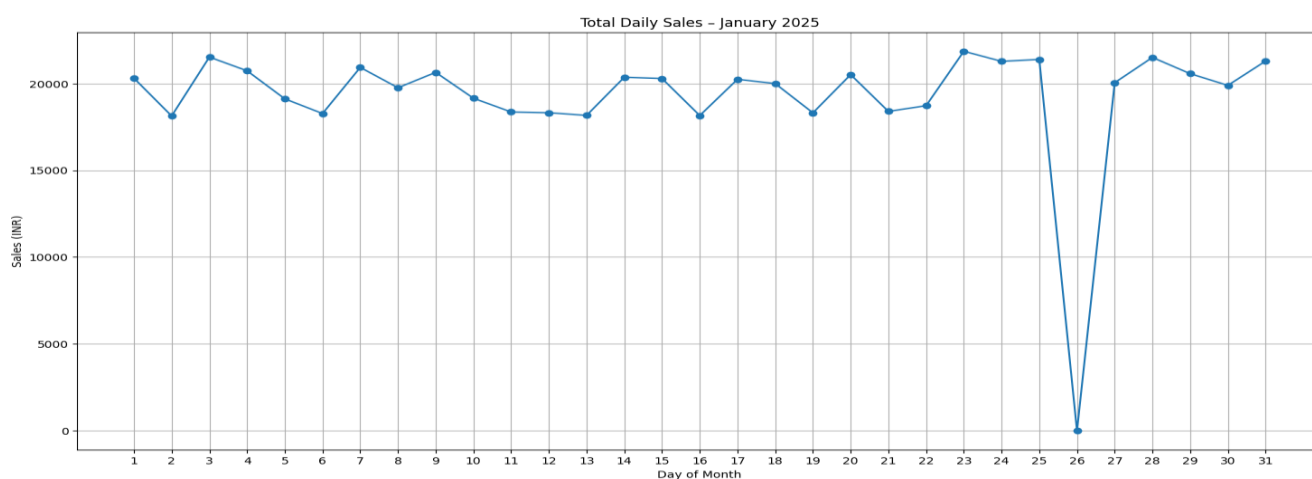
Descriptive Statistics	Amount (₹)
Sum	₹9,510.00
Mean	₹94.16
Median	₹80.00
Minimum price of product	₹10.00
Maximum price of product	₹500.00
Standard Deviation (Price Spread)	₹79.13
Total unique products	101
Total brands represented	70



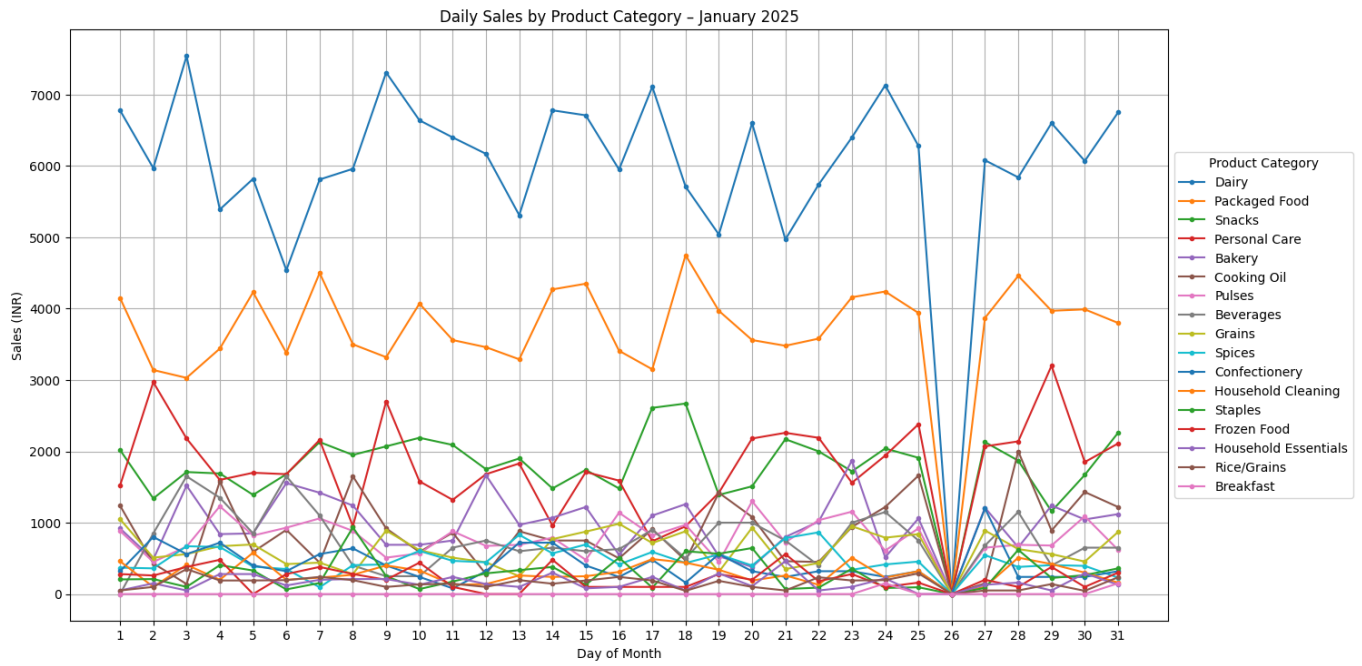
3.2.2 Sales Data:

Metric	Value
Total Units Sold (Month)	9,730
Total Revenue	₹596,005
Average Units Sold per Product	96.34
Average Revenue per Product	₹5,901.04

1a. Total Daily Sales: *Plotted between Days of Sales VS Daily Revenue*

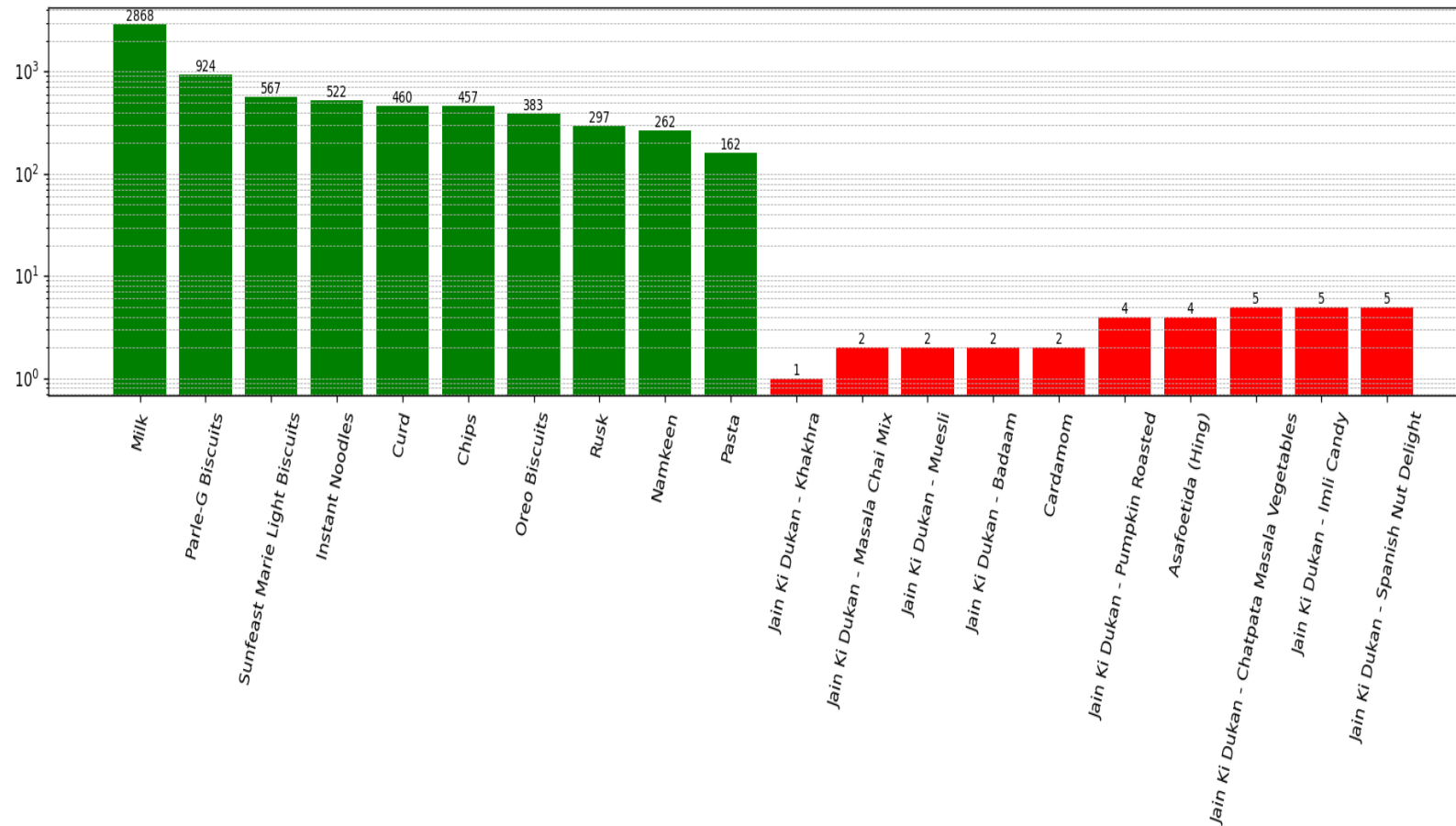


1b. Daily Sales by product: *Plotted between Daily Sales per product category VS Daily Revenue*

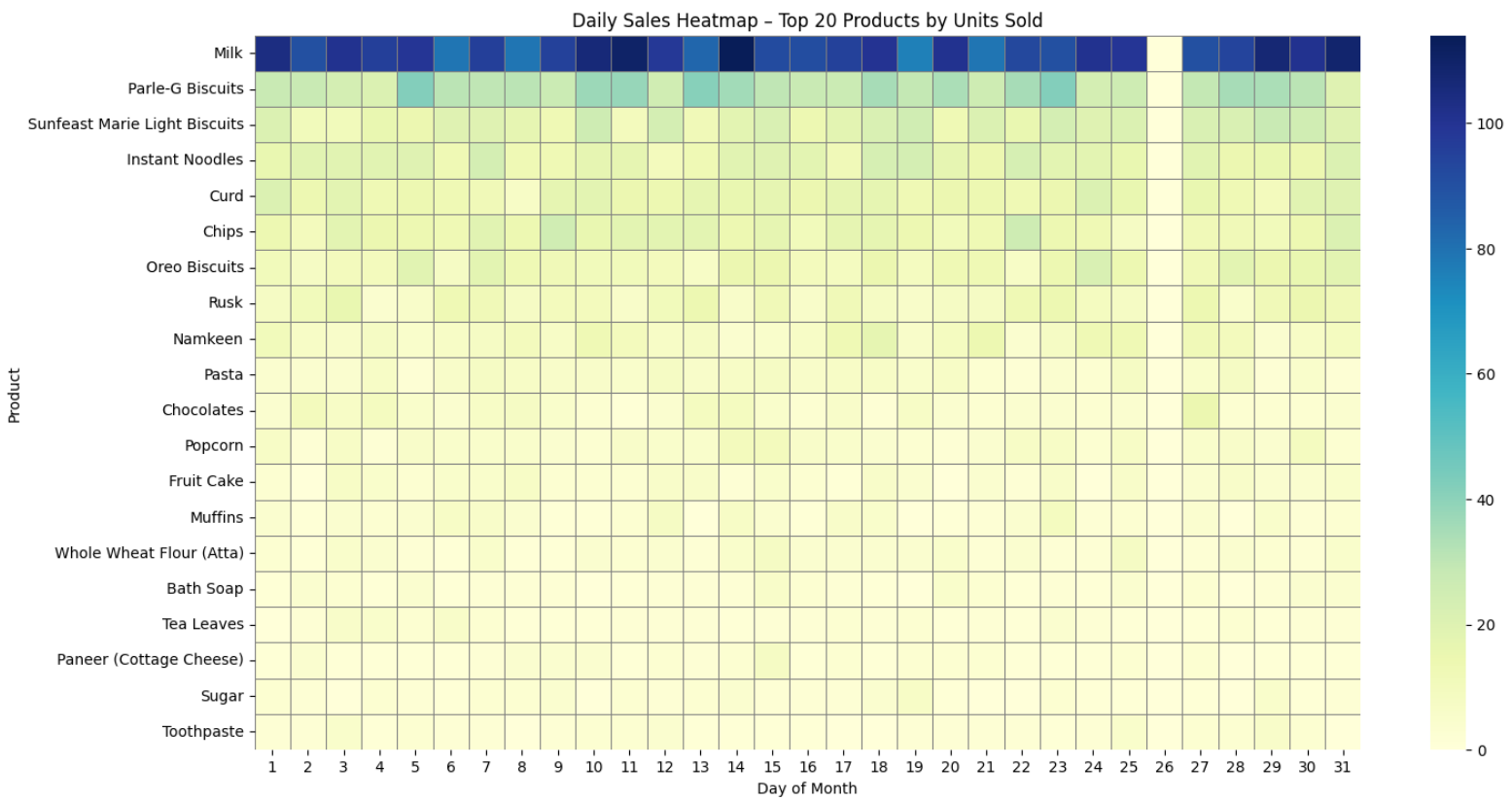


2a. Top 10 & Bottom 10 Products sold in a month (using log scale)

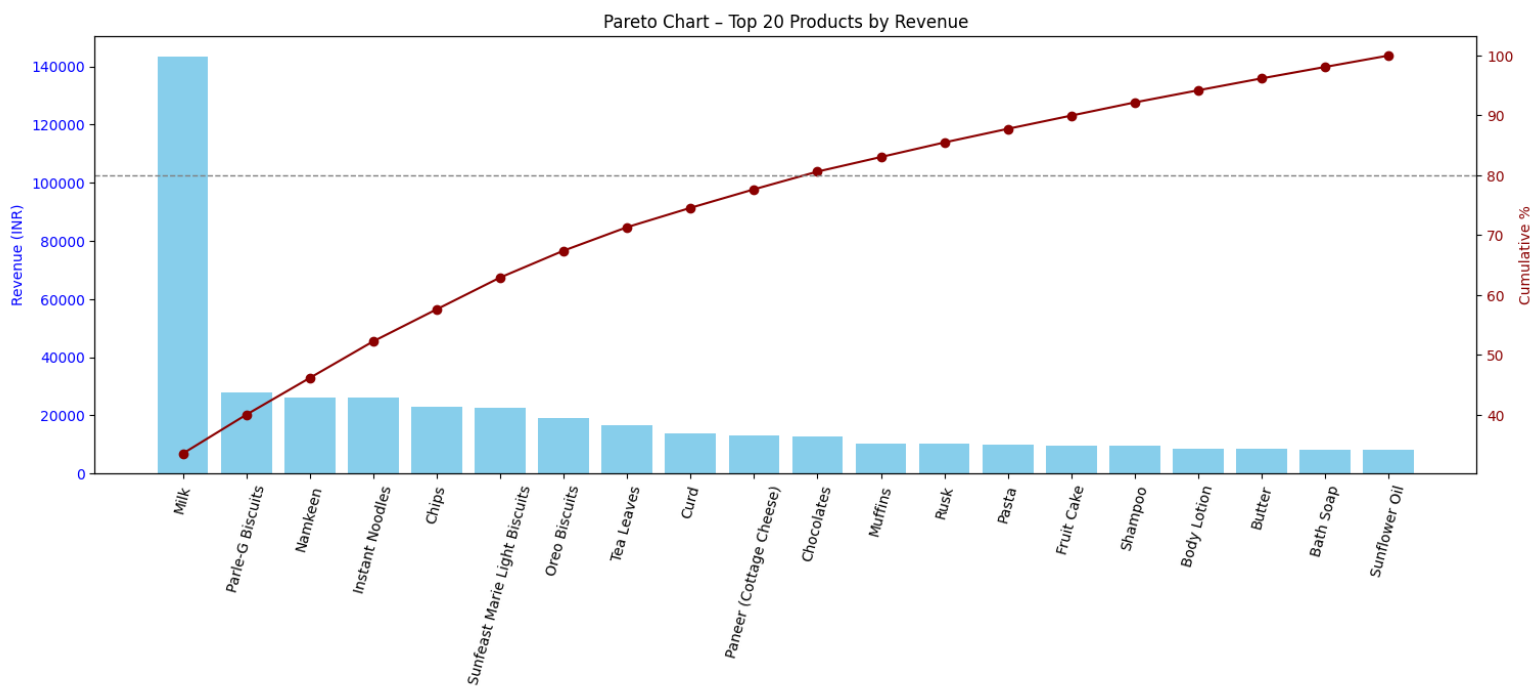
Top & Bottom 10 Products by Units Sold (Log Scale)



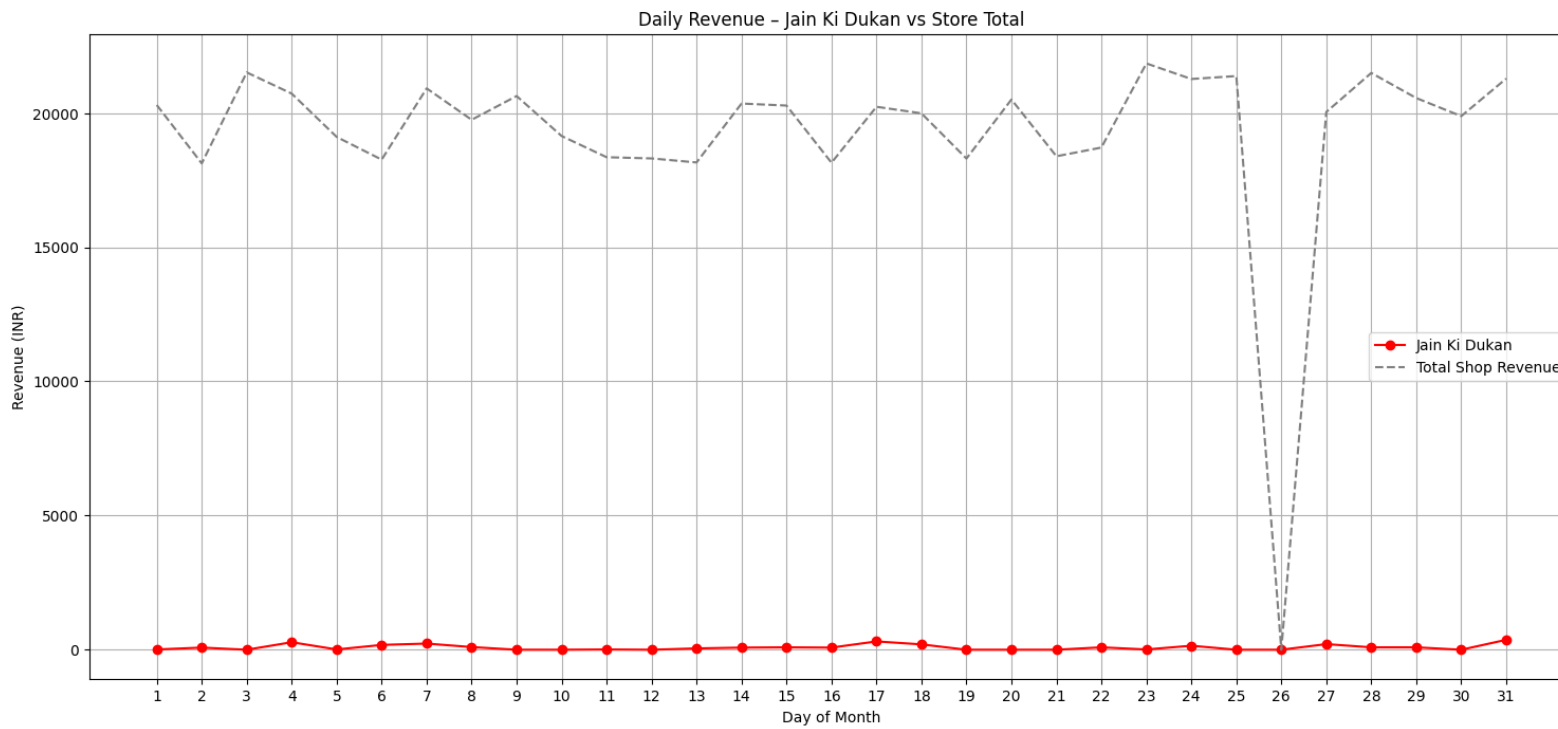
2b. Heatmap of top 20 products sold



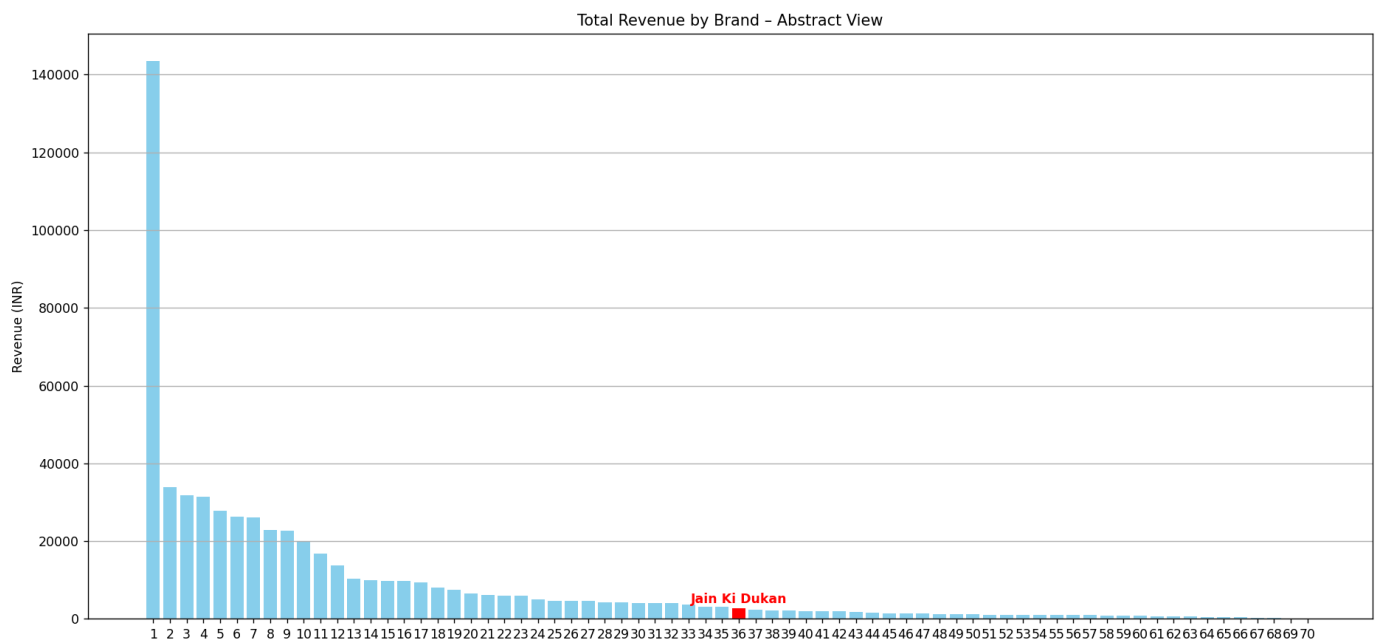
2c. Pareto Chart for top 20 products sold



3a. Comparing “Jain ki Dukan” items VS Other items on the basis of daily revenue



3b. Brand-wise monthly sales bar chart (Jain ki Dukan brand is highlighted in red)



4. Detailed explanation of Analysis Process and Methods:

The analysis process for the project involves accumulation of data and then using tools like Python and Excel, and concepts from BDM theory to figure out business problems. My main motive behind this was that the results I work towards should be meaningful to me and the owner, and not just devolve to senseless graphs, though I still tried to incorporate and learn new concepts.

Analysis was divided into 3 parts, as follows:

1. General monthly trends
2. Trends which will help us tackle problem 3 (mentioned in the proposal document) i.e. Inventory Management issues
3. Trends which will help us tackle problem 1 i.e. underperformance of self brand (Jain ki Dukan) products

Data collection: The first step of analysis process was to gather the data, but data was not maintained properly the shop, and often used manually inputting in excel sheets or use of physical notebooks. But this caused loss of data and problems with conversion of data to a suitable format for analysis. We figured out that the shop used Ezobill software for their billing and this can be converted to relevant formats ie CSV or Excel for each day separately. This proved to be really helpful in the analysis process.

Software used: Next onto the analysis part, once we had the data in excel format it was really easy to analyse it using BDM theory and data analysis skills learned in MLP(Machine Learning Practices) course.

I primarily used Python and a few handy functions in excel like SUM, UNIQUE etc. The choice was simple because of extensive libraries present in python like pandas and matplotlib. Pandas helped me convert the data into dataframes that can be easily manipulated like we do with excel sheets but this is faster as you don't have to manually do it.

Matplotlib is the main python library for plotting graphs which helped me visualise the graphs and then present them in my analysis.

Time-Series Analysis: This approach is particularly effective for retail and inventory data, which is inherently time-dependent. By analyzing daily sales trends, brand performance, and product category patterns over time, we gain valuable insights into the store's operational efficiency and financial health. This method is especially useful in identifying sales cycles, customer demand fluctuations, and the impact of specific events (like holidays). For instance, it helps us monitor underperforming brands such as *Jain Ki Dukan*, assess peak sales periods, and refine inventory and pricing strategies to maximize profitability and customer satisfaction.

Pareto Chart Analysis: We used a Pareto chart on the top 20 products to identify which items contributed most to revenue. Even within this group, a few products dominated sales, reflecting the 80/20 rule. This helped us focus on high-impact items for stocking and promotions, while also showing that the Jain Ki Dukan brand didn't appear among top performers—highlighting a need for strategic review.

Communication: Interestingly, communication with the shop owner and workers was one of the bottleneck in my project as I was not able to open up with the shop owner and it was difficult to convince them with radical new ideas. It was a bit easier once I talked to the owner's son who was around my age and understood the value of technology much more.

2. Results and Findings:

The following results and findings were found from the analysis processes:

1. General Monthly Trends

1a. Total Daily Sales:

The overall daily sales trend remained stable throughout January, with values ranging between ₹18,000 and ₹22,000 on most days. A significant anomaly appeared on the 26th, where sales dropped to nearly zero, suggesting a possible shop closure or data issue on that day.

1b. Daily Sales by Product Category:

Among all product categories, Dairy consistently outperformed others with high daily revenue, followed by Packaged Food and Snacks. The remaining categories showed relatively low and steady activity, indicating a clear disparity in sales distribution across product types.

2. Inventory Management Insights

2a. Top 10 & Bottom 10 Products Sold:

Using a log-scaled bar chart, the top-selling products, such as Milk and Parle-G Biscuits, were clearly distinguished from underperformers. Notably, most of the bottom 10 products—barely crossing even 5 units sold—were from the Jain Ki Dukan brand, indicating potential inventory waste and poor movement.

2b. Heatmap of Top 20 Products Sold:

The heatmap showed consistent daily sales for high-performing items, especially Milk and biscuits, which maintained regular demand. Conversely, several other products within the top 20 exhibited irregular patterns, suggesting possible stock-outs or inconsistent demand cycles.

2c. Pareto Chart for Top 20 Products Sold:

The Pareto chart revealed that a small subset of the top 20 products accounted for a disproportionate share of total revenue, aligning with the 80/20 principle. This insight is key for optimizing shelf space and focusing reordering strategies on high-impact products.

3. Self-Brand Performance

3a. Jain Ki Dukan vs Total Daily Revenue:

The comparative revenue chart illustrated a stark contrast between Jain Ki Dukan products and the store's total daily earnings. While the store generated ₹18,000–₹22,000 daily, Jain Ki Dukan's contributions were consistently under ₹1,000, indicating weak consumer traction.

3b. Brand-wise Monthly Sales:

A brand-wise revenue breakdown showed that Jain Ki Dukan ranked near the bottom among all brands carried by the store. Despite being an in-house brand, its poor performance compared to national brands points to issues in either product-market fit, pricing, or visibility.

In conclusion, the statistical analysis over a considerable period of data gave insights that could be used to reduce the problems faced by the shop. One solution that came across my mind during the analysis was to sell milk, their top performer, as their self brand product as it's clearly an outlier in their sales and performs really well due to daily consumption by the locals in the area.

3. Appendix:

